

ACTIVE EARTH

Earth's surface is constantly changing, thanks to dynamic geological processes driven by its internal heat energy. The plates of Earth's brittle outer crust are always in motion, altering the shape of oceans and continents as they do so.

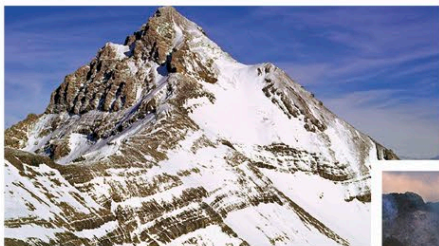
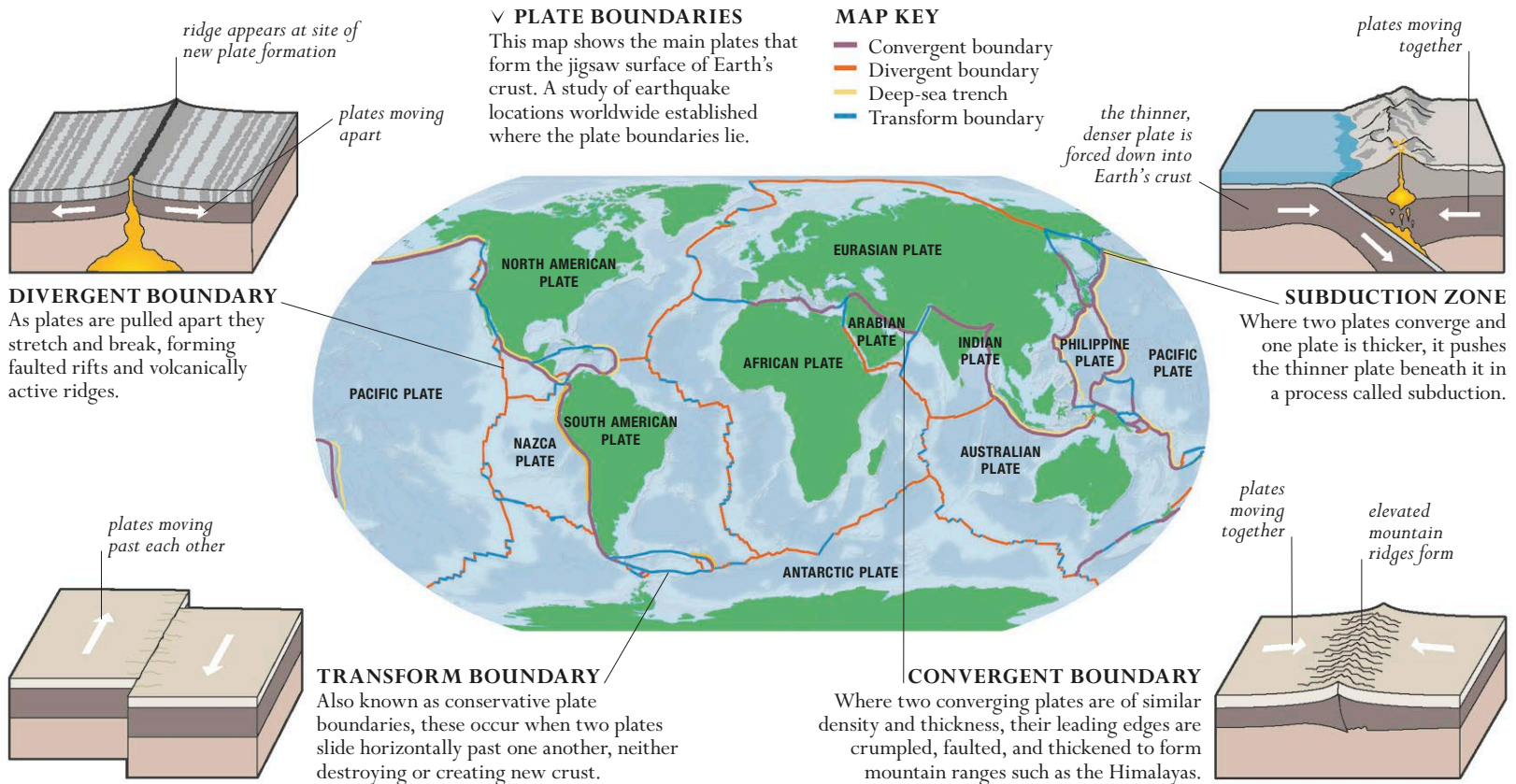
PLATE TECTONICS

Over geological time, Earth's surface – and the distribution and size of the continents and oceans – has constantly changed, driven by the process of plate tectonics. The cool and brittle rocks of Earth's outermost crust are broken into a number of semi-rigid slabs known as tectonic plates. There are seven major continent-sized plates and about a dozen smaller ones. Over time, crustal plates have jostled against one another. Hot magma rising from the mantle erupts from rifts in the crust to

form new ocean floor crust. This spreads away on either side of the rift to form divergent plate boundaries, mainly beneath the oceans. And, since Earth itself cannot expand, the creation of new oceanic crust requires that the crust is shortened elsewhere by the same amount. This reduction occurs at convergent boundaries where either one plate overrides the other – a process known as subduction – or the plate margins are compressed and buckled to form mountains.



▲ **SAN ANDREAS FAULT**
Stretching some 1,300 km (810 miles) through California, this dramatic fault is the product of a transform boundary between the Pacific and North American plates, which slide against one another.



< FOLD MOUNTAINS

The intense pressure caused by converging plate margins folds, faults, and thickens the rocks of the crust. And, crustal thickening elevates mountain ranges.



ACTIVE VOLCANOES >

Most volcanoes form at plate margins. Deep down, rocks melt to produce hot magma which rises and erupts at the surface. Even dormant volcanoes may one day erupt as plates shift beneath them.

MOUNTAINS AND VOLCANOES

One of the major factors controlling the movement and distribution of life on Earth is its varied topography – its surface features – including the towering mountains and volcanoes on land and under the sea. On land, mountains not only hinder the movement of wildlife but alter weather, climate, and local plant life, which in turn impacts upon animal life. Active volcanoes also affect their surroundings when they erupt – initially by destroying life, but also, in the longer term, when the weathering and erosion of erupted lava and ash provide new mineral nutrients that fertilize the area. Submarine mountains and volcanic eruptions affect the movement of water, marine life, and the fertility of ocean waters.